

Scattering Amplitudes Without Diagrams CHY and Ambitwistor Strings

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$$n \quad 1$$

$$\text{LSZ} \implies \boxed{\mathcal{A}_n = \text{Res}(G(P_i))|_{P_a^2=0}}$$

$$S \Rightarrow \int \mathcal{D}\phi e^{iS[\phi]} \Rightarrow \text{diagram with a horizontal line and four external legs} + \dots \Rightarrow \frac{1}{s_{12} - m^2} + \dots = \mathcal{A}_n$$

Why look for something new?

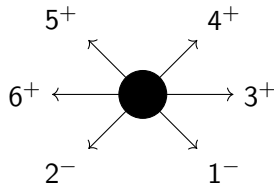
Traditional methods:

- ▷ **Combinatorial explosion** of diagrams.
- ▷ **Loop integrals** become **complicated**.

But the final answers look...simple!

1986 – Parke–Taylor [26]

$$\mathcal{A}_n^{\text{MHV}}(i^-, j^-) = i \frac{\langle ij \rangle^4}{\prod_{a=1}^n \langle a \ a+1 \rangle}$$

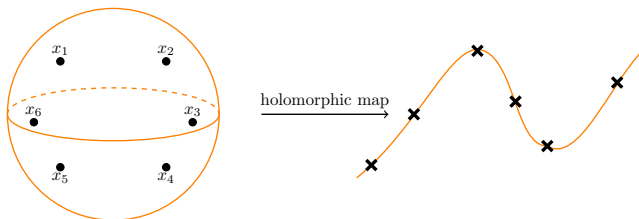


Hint of a Hidden Structure

1988 - Nair [25]: MHV amplitudes $\leftrightarrow$ current correlators on $\mathbb{CP}^1$.

2003 - Witten [35]: Twistor string on $\mathbb{CP}^{3|4}$ encodes $\mathcal{N} = 4$ SYM amplitudes.

2004 - RSV [30]: Tree amplitudes = integrals over moduli spaces of holomorphic maps between:



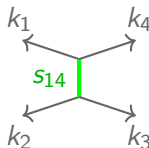
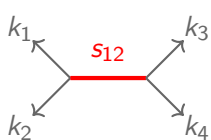
Can we break from the perturbative story?

The CHY Scattering Map

CHY: Amplitudes computed directly from kinematic data.

Kinematics $\leftrightarrow$ Geometry: No Diagrams Needed!

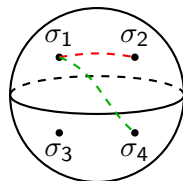
$$s_{ab} = (k_a + k_b)^2 \quad \longleftrightarrow \quad \sigma_a \in \mathbb{CP}^1$$



Scattering Map



$$\{k_i\} \mapsto \{\sigma_i\}$$



$$\mathcal{M}_{0,4} \ni (\sigma_1, \dots, \sigma_4)$$

▷ How does this map come about?

The CHY Scattering Equations

The Scattering Equations [10, 8, 11, 12, 9]

For n massless particles, the $n-3$ *scattering equations* are

$$\sum_{\substack{b=1 \\ b \neq a}}^n \frac{s_{ab}}{\sigma_a - \sigma_b} = 0, \quad \text{for all } a = 1, \dots, n \quad (1)$$

which have $(n-3)!$ solutions.

- ▷ They form the backbone of the tree-level S-matrix for massless particles in **any** dimension. [8, 12]

Tree-Level, Massless Amplitudes* in CHY

Universal CHY Representation [12, 11]

$$A_n^{\text{massless}} = \frac{1}{\text{Vol}(\text{SL}(2, \mathbb{C}))} \int \prod_{a=1}^n d\sigma_a \prod'_a \delta \left(\sum_{b \neq a} \frac{s_{ab}}{\sigma_a - \sigma_b} \right) \mathcal{I}_L \mathcal{I}_R$$

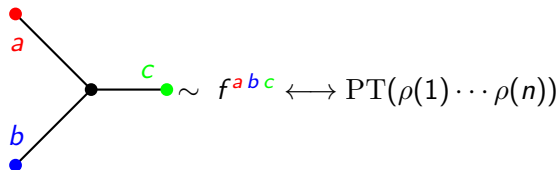
- ▷ **Universal** for a wide class of tree-level, massless theories [22].
- ▷ $\frac{1}{\text{Vol}(\text{SL}(2, \mathbb{C}))}$: gauge fixing $\text{SL}(2, \mathbb{C})$ redundancy of $\mathbb{CP}^1$.
- ▷ $\mathcal{I}_L, \mathcal{I}_R$: **theory-dependent half-integrands**.

*For theories, like Yang-Mills & the Bi-adjoint Scalar Theory, with colour or flavour symmetry, this is just the *partial* or *double-partial* amplitude.

The Half Integrands - Part 1: Colour

Theories like Yang-Mills have *colour* symmetry:

$$\mathcal{L} = -\frac{1}{4} \sum_a F_{\mu\nu}^a F^{a,\mu\nu}, \quad F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c.$$



$$PT(\rho(1) \cdots \rho(n)) \equiv \frac{1}{(\sigma_{\rho(1)} - \sigma_{\rho(2)}) \cdots (\sigma_{\rho(n)} - \sigma_{\rho(1)})}$$

Colour ordering $\leftrightarrow$ *Cyclic Ordering of punctures on the Sphere*

The Half Integrands - Part 2: Encoding Polarisation

- ▷ Polarisation is encoded by the *reduced Pfaffian* of the *antisymmetric* matrix $\text{Pf}' M$:

$$M_{2n} = \begin{bmatrix} A & -C^T \\ C & B \end{bmatrix},$$

where

$$A_{ab} := \frac{k_a \cdot k_b}{x_a - x_b}, \quad B_{ab} := \frac{\epsilon_a \cdot \epsilon_b}{x_a - x_b},$$

$$C_{aa} := - \sum_{\substack{b=1 \\ b \neq a}}^n C_{ab} \quad \text{and} \quad C_{ab} := \frac{\epsilon_a \cdot k_b}{x_a - x_b}.$$

- ▷ $(\text{Pf}' M)^2 = \det(M)$
- ▷ The *reduced Pfaffian* = eliminate two rows and two columns (dimensional analysis and $\text{Pf} M = 0$).

The Double & Zero Copy

Spin	QFT	$\mathcal{I}_L$	$\mathcal{I}_R$
$s = 0$	Bi-adjoint Scalar (BA ϕ^3)	$\text{PT}(\alpha)$	$\text{PT}(\beta)$
$s = 1$	YM	$\text{PT}(\alpha)$	$\text{Pf}'(M)$
$s = 2$	Gravity	$\text{Pf}'(M)$	$\text{Pf}'(\tilde{M})$

Using a very nice property of the scattering equations solutions called *KLT orthogonality*, one can show:

- ▷ **Zeroth Copy:** Replace $\text{PT}(\beta)$ in BA_{ϕ^3} with $\text{Pf}'(M) = \text{YM}$.
- ▷ **Double Copy:** $\text{Gravity} = \text{YM} \otimes_{\text{KLT}} \text{YM}$

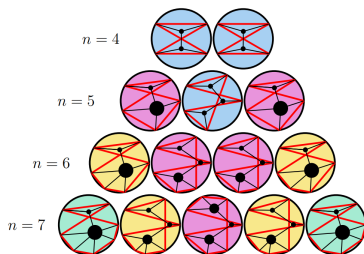
What I am Happy to Chat About Later!

What I Have Been/Am Exploring - Bi-adjoint Scalar Theory

Massive Scalars

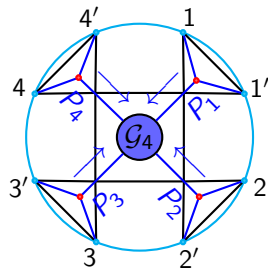
$$\mathcal{S}(\{\sigma_i\}) = \sum_{k=2}^{n-2} \sum_{i=1}^k (s_{i,\dots,i+n-k-1} - m^2) \log \left(\frac{|\sigma_{i,i+n-k-1}| |\sigma_{i+n-k,i-1}|}{|\sigma_{i,i+n-k}| |\sigma_{i+n-k-1,i-1}|} \right)$$

Combinatorics



2604.24830 [32]

Off-Shell Regime/ $s_{ij} = 0$

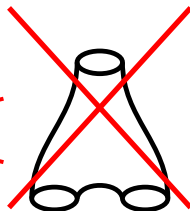


Where do they come from?!

CHY looks *nothing* like:

- ▷ a space-time action
- ▷ conventional string theory

~~$$S[\phi] = \int d^4x \mathcal{L}$$~~



So what is the underlying physics?



Ambitwistor Strings

Ambitwistor Space [18, 1]

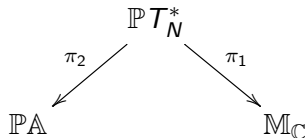
1. Start with $\mathbb{M}_{\mathbb{C}}$ in d -dimensions.
2. Cotangent Bundle of $\mathbb{M}_{\mathbb{C}}$: $(X^a, P_a) \in T^*\mathbb{M}_{\mathbb{C}}$
3. Space of Null Directions: $T_N^* = \{(X, P) \in T^*\mathbb{M}_{\mathbb{C}} | P^2 = 0\}$
4. **Mod out by shifts in spacetime along each null direction:**

$$\mathbb{A} = T_N^* / \left\{ P \cdot \frac{\partial}{\partial X} \right\}$$

5. However, we only want the **projective** information of the space:

Projective Ambitwistor Space

$$\mathbb{PA} = \mathbb{A} / \left\{ P \cdot \frac{\partial}{\partial P} \right\}$$



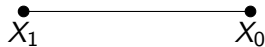
Worldline Quantum Field Theory

Amplitudes are now path integrals over *particle trajectories*.

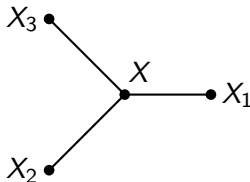
$$D_F(X_1, X_0) = \int \frac{d^d p}{(2\pi)^d} \frac{e^{ip \cdot (x_1 - x_0)}}{p^2}$$

$$= \int_0^\infty dT \int \frac{d^d p}{(2\pi)^d} e^{-Tp^2 + ip \cdot (X_1 - X_0)}$$

$$= \int_0^\infty dT \int_{X(0)=X_0}^{X(T)=X_1} \mathcal{D}X \exp \left(-\frac{1}{2} \int_0^T \dot{X}^2 d\tau \right)$$



Feynman Diagrams = Worldlines Glued Together



$$= \int d^3 T \int d^d X e^{-S_\Gamma[X]}$$

$$= \int d^d x \prod_{i=1}^3 \int \frac{d^d p_i}{(2\pi)^d} \frac{e^{ip_i \cdot (X_i - X)}}{p_i^2}$$

$$= \langle \phi(x_1) \phi(x_2) \phi(x_3) \rangle_{\text{tree}}$$

First Order Action

$$S[X, P] = \int P_\mu dX^\mu - \frac{1}{2} \tilde{e} g^{\mu\nu} P_\mu P_\nu,$$

▷ $\tilde{e}$ enforces $P^2 = 0$ from its e.o.m.

Gauge Transformations

$$\delta X^\mu = \alpha g^{\mu\nu} P_\nu, \quad \delta P_\mu = 0, \quad \delta \tilde{e} = d\alpha$$

▷ Gauge transformations send fields X to X' along the null translation generated by P .

Solutions to the field equations *modulo gauge* are **null geodesics** in spacetime, parameterised by the scaling of P .

Ambitwistor Strings: Definition & Algorithm [18]

QFTs of holomorphic maps from Riemann surfaces to $\mathbb{PA}$.

1. **Worldline** $(\tau) \longrightarrow$ **Worksheet** (Σ, σ)

2. **Complexify space-time:**

$$(X^\mu, P_\mu) \in T^*\mathbb{M}_{\mathbb{C}}, \quad \text{holomorphic}$$

3. **Holomorphicity:** $dX \rightarrow \bar{\partial}_e X = d\bar{\sigma} \partial_{\bar{\sigma}} X - e \partial_{\sigma} X$; $e \in \Omega^{0,1}(\Sigma, T_{\Sigma})$.

$$S_{\text{bos}}[X, P] = \frac{1}{2\pi} \int_{\Sigma} P_{\mu} \bar{\partial}_e X^{\mu} - \frac{\tilde{e}}{2} g^{\mu\nu} P_{\mu} P_{\nu}$$

4. **Gauge Transformations:** $\delta \tilde{e} = d\alpha, \longrightarrow \delta \tilde{e} = \bar{\partial} \alpha,$

5. $P^2 = 0$ via $\tilde{e}$ e.o.m $\implies T_N^* \longrightarrow$ **Ambitwistor Space**

Ingredients of an Ambitwistor String [18]

All Models Take the Form

$$S = S_{\text{bos.}} + S_L + S_R$$

1. **Current Algebras** S_C : 'currents' $j^a \in K_\Sigma \otimes \mathfrak{g}$

$$j^a(\sigma)j^b(\sigma') \sim \frac{f_c^{ab}j^c}{\sigma - \sigma'} + \frac{k\delta^{ab}}{(\sigma - \sigma')^2}$$

2. **Worldsheet Superalgebra**: fermionic spinor $\Psi^\mu \in K^{1/2} \otimes \mathbb{C}^d$ & fermionic gauge field $\chi \in \Omega^{0,1} \otimes T_\Sigma^{1/2}$

$$S_\Psi = \int g_{\mu\nu} \Psi^\mu \bar{\partial} \Psi^\nu - \chi P \cdot \Psi.$$

Ambitwistor String Structures $\longleftrightarrow$ CHY Structures [18]

What worldsheet matter corresponds to $\mathcal{I}_L$ and $\mathcal{I}_R$ of CHY?

QFT	Ambitwistor String	CHY
Biadjoint Scalar	$S_{C_\alpha} + S_{\tilde{C}_\beta}$	$\text{PT}(\alpha) \text{PT}(\beta)$
YM	$S_\Psi + S_{C_\alpha}$	$\text{PT}(\alpha) \text{Pf}'(M)$
Gravity	$S_\Psi + S_{\tilde{\Psi}}$	$\text{Pf}'(M) \text{Pf}'(\tilde{M})$

How do CHY amplitudes arise?

- ▷ Insert vertex operators impose constraints (via $\delta(k \cdot P)$).
- ▷ Compute their correlator:

$$\mathcal{A}_n = \langle \mathcal{V}_1 \cdots \mathcal{V}_n \rangle$$

- ▷ The path integral localises onto the scattering equations.

Applications & Breakthroughs Due to Ambitwistor Strings

Twistors give a geometric description of amplitudes in **4d**. Ambitwistor strings extend this to ***all dimensions***, where **scattering equations** arise from a ***universal worldsheet theory***.

- ▷ **Loop amplitudes:** scattering equations & double copy beyond tree level e.g. [4, 19, 20, 21]
- ▷ **Massive extensions:** double copy structures, masses $\leftrightarrow$ internal symmetry group etc. [5, 6].
- ▷ **Curved backgrounds:** beyond flat spacetime [3, 2]

Thank you for Listening!

Any questions?

References I

- [1] Tim Adamo. *Lectures on twistor theory*. 2018. arXiv: 1712.02196 [hep-th]. URL: <https://arxiv.org/abs/1712.02196>.
- [2] Tim Adamo, Eduardo Casali, and Stefan Nekovar. “Ambitwistor string vertex operators on curved backgrounds”. In: *Journal of High Energy Physics* 2019.1 (Jan. 2019). ISSN: 1029-8479. DOI: [10.1007/jhep01\(2019\)213](https://doi.org/10.1007/jhep01(2019)213). URL: [http://dx.doi.org/10.1007/JHEP01\(2019\)213](http://dx.doi.org/10.1007/JHEP01(2019)213).

References II

- [3] Tim Adamo, Eduardo Casali, and David Skinner. “A worldsheet theory for supergravity”. In: *Journal of High Energy Physics* 2015.2 (Feb. 2015). ISSN: 1029-8479. DOI: 10.1007/jhep02(2015)116. URL: [http://dx.doi.org/10.1007/JHEP02\(2015\)116](http://dx.doi.org/10.1007/JHEP02(2015)116).
- [4] Tim Adamo, Eduardo Casali, and David Skinner. “Ambitwistor strings and the scattering equations at one loop”. In: *Journal of High Energy Physics* 2014.4 (Apr. 2014). ISSN: 1029-8479. DOI: 10.1007/jhep04(2014)104. URL: [http://dx.doi.org/10.1007/JHEP04\(2014\)104](http://dx.doi.org/10.1007/JHEP04(2014)104).

References III

- [5] Giulia Albonico, Yvonne Geyer, and Lionel Mason. “From Twistor-Particle Models to Massive Amplitudes”. In: *SIGMA* 18 (2022), p. 045. DOI: 10.3842/SIGMA.2022.045. arXiv: 2203.08087 [hep-th].
- [6] Giulia Albonico, Yvonne Geyer, and Lionel Mason. “Massive ambitwistor-strings; twistorial models”. In: *JHEP* 01 (2024), p. 127. DOI: 10.1007/JHEP01(2024)127. arXiv: 2301.11227 [hep-th].
- [7] Freddy Cachazo. *PSI 2017/2018 - Scattering Amplitudes in QFT & String Theory - Lecture 1*. en. PIRSA:18040035 see, <https://pirsa.org>. 2018. URL: <https://pirsa.org/18040035>.

References IV

- [8] Freddy Cachazo, Song He, and Ellis Ye Yuan. “Scattering equations and Kawai-Lewellen-Tye orthogonality”. In: *Physical Review D* 90.6 (Sept. 2014). ISSN: 1550-2368. DOI: [10.1103/physrevd.90.065001](https://doi.org/10.1103/physrevd.90.065001). URL: <http://dx.doi.org/10.1103/PhysRevD.90.065001>.
- [9] Freddy Cachazo, Song He, and Ellis Ye Yuan. “Scattering equations and matrices: from Einstein to Yang-Mills, DBI and NLSM”. In: *Journal of High Energy Physics* 2015.7 (July 2015). ISSN: 1029-8479. DOI: [10.1007/jhep07\(2015\)149](https://doi.org/10.1007/jhep07(2015)149). URL: [http://dx.doi.org/10.1007/JHEP07\(2015\)149](http://dx.doi.org/10.1007/JHEP07(2015)149).

References V

- [10] Freddy Cachazo, Song He, and Ellis Ye Yuan. “Scattering in three dimensions from rational maps”. In: *Journal of High Energy Physics* 2013.10 (Oct. 2013). ISSN: 1029-8479. DOI: 10.1007/jhep10(2013)141. URL: [http://dx.doi.org/10.1007/JHEP10\(2013\)141](http://dx.doi.org/10.1007/JHEP10(2013)141).
- [11] Freddy Cachazo, Song He, and Ellis Ye Yuan. “Scattering of Massless Particles in Arbitrary Dimensions”. In: *Physical Review Letters* 113.17 (Oct. 2014). ISSN: 1079-7114. DOI: 10.1103/physrevlett.113.171601. URL: <http://dx.doi.org/10.1103/PhysRevLett.113.171601>.

References VI

- [12] Freddy Cachazo, Song He, and Ellis Ye Yuan. “Scattering of massless particles: scalars, gluons and gravitons”. In: *Journal of High Energy Physics* 2014.7 (July 2014). ISSN: 1029-8479. DOI: 10.1007/jhep07(2014)033. URL: [http://dx.doi.org/10.1007/JHEP07\(2014\)033](http://dx.doi.org/10.1007/JHEP07(2014)033).
- [13] Freddy Cachazo, Sebastian Mizera, and Guojun Zhang. “Scattering equations: real solutions and particles on a line”. In: *Journal of High Energy Physics* 2017.3 (Mar. 2017). ISSN: 1029-8479. DOI: 10.1007/jhep03(2017)151. URL: [http://dx.doi.org/10.1007/JHEP03\(2017\)151](http://dx.doi.org/10.1007/JHEP03(2017)151).
- [14] Lance J Dixon. *A brief introduction to modern amplitude methods*. en. 2014. DOI: 10.5170/CERN-2014-008.31. URL: <https://cds.cern.ch/record/1613349>.

References VII

- [15] Lance J. Dixon. “Calculating scattering amplitudes efficiently”. In: *Theoretical Advanced Study Institute in Elementary Particle Physics (TASI 95): QCD and Beyond*. Jan. 1996, pp. 539–584. arXiv: hep-ph/9601359.
- [16] Louise Dolan and Peter Goddard. “Proof of the formula of Cachazo, He and Yuan for Yang-Mills tree amplitudes in arbitrary dimension”. In: *Journal of High Energy Physics* 2014.5 (May 2014). ISSN: 1029-8479. DOI: 10.1007/jhep05(2014)010. URL: [http://dx.doi.org/10.1007/JHEP05\(2014\)010](http://dx.doi.org/10.1007/JHEP05(2014)010).
- [17] Henriette Elvang and Yu-tin Huang. “Scattering Amplitudes”. In: (Aug. 2013). arXiv: 1308.1697 [hep-th].

References VIII

- [18] Yvonne Geyer and Lionel Mason. *The SAGEX Review on Scattering Amplitudes, Chapter 6: Ambitwistor Strings and Amplitudes from the Worldsheet*. 2022. arXiv: 2203.13017 [hep-th]. URL: <https://arxiv.org/abs/2203.13017>.
- [19] Yvonne Geyer et al. “Loop Integrands for Scattering Amplitudes from the Riemann Sphere”. In: *Phys. Rev. Lett.* 115.12 (2015), p. 121603. DOI: 10.1103/PhysRevLett.115.121603. arXiv: 1507.00321 [hep-th].
- [20] Yvonne Geyer et al. “One-loop amplitudes on the Riemann sphere”. In: *JHEP* 03 (2016), p. 114. DOI: 10.1007/JHEP03(2016)114. arXiv: 1511.06315 [hep-th].

References IX

- [21] Yvonne Geyer et al. “Two-Loop Scattering Amplitudes from the Riemann Sphere”. In: *Phys. Rev. D* 94.12 (2016), p. 125029. DOI: 10.1103/PhysRevD.94.125029. arXiv: 1607.08887 [hep-th].
- [22] Zurab Jashi, Jaroslav Scheinpflug, and Yale Yauk. *Scaffolding Residues in Yang-Mills-Scalar à la CHY*. 2024. arXiv: 2411.12807 [hep-th]. URL: <https://arxiv.org/abs/2411.12807>.
- [23] H. Lehmann, K. Symanzik, and W. Zimmermann. “On the formulation of quantized field theories”. In: *Nuovo Cim.* 1 (1955), pp. 1425–1452. DOI: 10.1007/BF02731765.

References X

- [24] Michelangelo L. Mangano and Stephen J. Parke. “Multiparton amplitudes in gauge theories”. In: *Phys. Rept.* 200 (1991), pp. 301–367. DOI: 10.1016/0370-1573(91)90091-Y. arXiv: hep-th/0509223.
- [25] V. P. Nair. “A Current Algebra for Some Gauge Theory Amplitudes”. In: *Phys. Lett. B* 214 (1988), pp. 215–218. DOI: 10.1016/0370-2693(88)91471-2.
- [26] Stephen J. Parke and T. R. Taylor. “An Amplitude for n Gluon Scattering”. In: *Phys. Rev. Lett.* 56 (1986), p. 2459. DOI: 10.1103/PhysRevLett.56.2459.
- [27] R. Penrose. “Twistor algebra”. In: *J. Math. Phys.* 8 (1967), p. 345. DOI: 10.1063/1.1705200.

References XI

- [28] M.E. Peskin. *Concepts of Elementary Particle Physics*. Oxford Master Series in Condensed Matter Physics Series. Oxford University Press, 2019. ISBN: 9780198812180. URL: <https://books.google.ca/books?id=BpGqDwAAQBAJ>.
- [29] T.K. Petersen. *Eulerian Numbers*. Birkhäuser Advanced Texts Basler Lehrbücher. Springer New York, 2016. ISBN: 9781493947942. URL: <https://books.google.ca/books?id=V42WtAEACAAJ>.
- [30] Radu Roiban, Marcus Spradlin, and Anastasia Volovich. “Tree-level Smatrix of Yang-Mills theory”. In: *Physical Review D* 70.2 (July 2004). ISSN: 1550-2368. DOI: 10.1103/physrevd.70.026009. URL: <http://dx.doi.org/10.1103/PhysRevD.70.026009>.

References XII

- [31] J. A. Segner. “Enumeratio modorum quibus figurae planae ...”. Latin. In: *Novi Commentarii Academiae Scientiarum Imperialis Petropolitanae* 7 (1761). Dated 1758/59, published in 1761, pp. 203–210.
- [32] Lauren Smyth. *Emergent Features in $U(N) \times U(\tilde{N})$ Bi-adjoint Cubic Theory*. 2026. arXiv: 2604.24830 [hep-th]. URL: <https://arxiv.org/abs/2604.24830>.
- [33] Richard P. Stanley. *Enumerative Combinatorics, Volume 1*. 2nd. See Chapter 6 for Dyck paths and Catalan structures. Cambridge University Press, 2011.
- [34] Mark Thomson. *Modern Particle Physics*. Cambridge University Press, 2013.

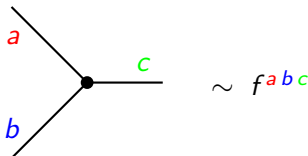
References XIII

- [35] Edward Witten. “Perturbative Gauge Theory as a String Theory in Twistor Space”. In: *Communications in Mathematical Physics* 252.1–3 (Oct. 2004), pp. 189–258. ISSN: 1432-0916. DOI: [10.1007/s00220-004-1187-3](https://doi.org/10.1007/s00220-004-1187-3). URL: <http://dx.doi.org/10.1007/s00220-004-1187-3>.

A Quick Colour-Ordering Tutorial

Yang-Mills Theory:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^a F^{a,\mu\nu}, \quad F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc}A_\mu^b A_\nu^c.$$



$$T^{a_i} \in U(N)$$

$$i[T^{a_1}, T^{a_2}] = \sum_{a_3=1}^{N^2} f^{a_1 a_2 a_3} T^{a_3}$$

$$f^{a_1 a_2 a_3} = \text{tr}(T^{a_1} T^{a_2} T^{a_3}) - \text{tr}(T^{a_2} T^{a_1} T^{a_3}).$$

Colour Decomposition

All tree-level amplitudes can be written as

$$\mathcal{A}_n = \sum_{\rho \in S_n / \mathbb{Z}_n} \text{tr}(T^{a_{\rho_1}} \dots T^{a_{\rho_n}}) A_n(\rho(1) \dots \rho(n)),$$

where A_n is the *partial amplitude*.

CHY Amplitudes & A Colour-Ordered Scalar Theory

Adding Colour to CHY [11]

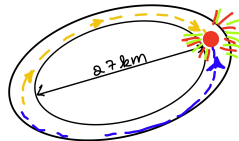
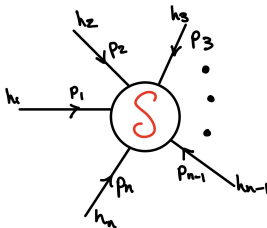
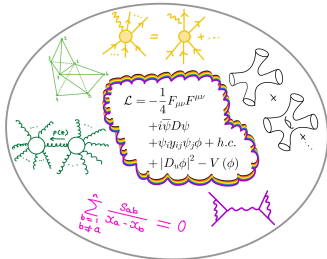
Colour-dressed integrands $\tilde{\mathcal{I}}_L$ & $\tilde{\mathcal{I}}_R$ for Yang-Mills ($s = 1$) & Einstein gravity ($s = 2$):

$$\tilde{\mathcal{I}}_L \tilde{\mathcal{I}}_R = \left(\sum_{\rho \in S_n / \mathbb{Z}_n} \frac{\text{Tr}(T^{a_{\rho(1)}} \dots T^{a_{\rho(n)}})}{(x_{\rho(1)} - x_{\rho(2)}) \dots (x_{\rho(n)} - x_{\rho(1)})} \right)^{2-s} (\text{Pf}' M)^s. \quad (2)$$

But what about for $s = 0$?

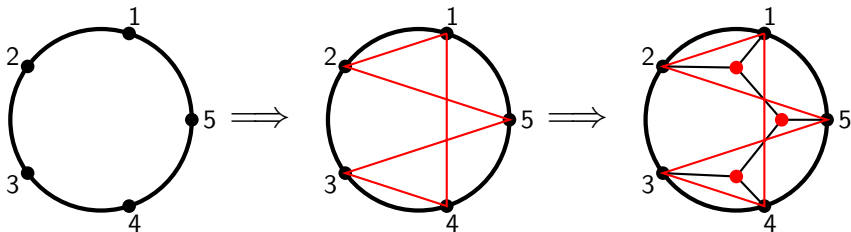
$\implies$ A scalar theory with two independent colour-orderings?!

What are we all here for?



Geometric Structure of Bi-adjoint ϕ^3 Amplitudes

How to Draw a Double Partial Amplitude: $m_5(12345|12534)$



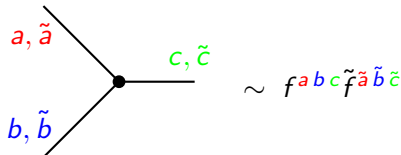
- ▷ Only planar diagrams allowed!
- ▷ # n -particle diagrams = triangulations of $n - 2$ polygon = C_{n-2} .
- ▷ Field theory behind the associahedron!

Tree-Level, Massless Bi-adjoint ϕ^3 Theory CHY AmplitudeInteractions in Bi-adjoint ϕ^3 Theory: $U(N) \times U(\tilde{N})$ symmetry

$$\mathcal{L}_{\text{int.}}^{\text{BA-}\phi^3} = -f_{abc} \tilde{f}_{\tilde{a}\tilde{b}\tilde{c}} \phi^{a\tilde{a}} \phi^{b\tilde{b}} \phi^{c\tilde{c}}. \quad (3)$$

▷ $U(N)$ generators as basis:

$$i[T^a, T^b] = \sum_{c=1}^{N^2} f^{abc} T^c.$$



▷ Split amplitude into colour and kinematic parts[12]:

$$\mathcal{A}_n^{\phi^3}(\{k_i, a_i, \tilde{a}_i\}) = \sum_{\alpha, \beta \in S_n / \mathbb{Z}_n} \text{Tr}(T^{a_{\alpha(1)}} \dots T^{a_{\alpha(n)}}) \text{Tr}(\tilde{T}^{\tilde{a}_{\beta(1)}} \dots \tilde{T}^{\tilde{a}_{\beta(n)}}) m_n(\alpha|\beta).$$

$$\triangleright \mathcal{I}_L^{\text{BA } \phi^3} = \frac{1}{(x_{\alpha(1)} - x_{\alpha(2)}) \dots (x_{\alpha(n)} - x_{\alpha(1)})}$$

$$\triangleright \mathcal{I}_R^{\text{BA } \phi^3} = \frac{1}{(x_{\beta(1)} - x_{\beta(2)}) \dots (x_{\beta(n)} - x_{\beta(1)})}$$

Extra Slides

A Massive Planar Scattering Potential

Combinatorial Structures & Cross-Ratios

Ordered Massive Particles - Doland & Goddard's Construction

Doland & Goddard's Massive Extension to the Scattering Eq.s [16]

For a ϕ^3 theory describing scalar particles of mass m , consider the following modification to the scattering equations:

$$\sum_{\substack{b \neq a \\ b \in A}} \frac{k_a \cdot k_b}{x_a - x_b} + \frac{m^2}{2(x_a - x_{a-1})} + \frac{m^2}{2(x_a - x_{a+1})} = 0, \quad (4)$$

where $A = \{1, 2, \dots, n\}$.

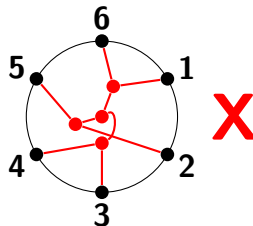
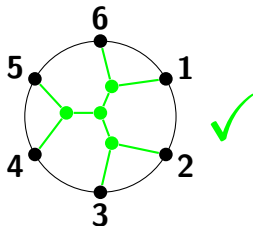
- ▷ Correctly reproduces tree amplitudes, but massive particles now have a **cyclical ordering**.
- ▷ Great that this works...but why?

Planar Feynman Diagrams

Colour-Ordering Origin

Connects to colour-ordered amplitudes in the bi-adjoint ϕ^3 theory.

▷ Graphically $\implies$ Planar Feynman Diagrams



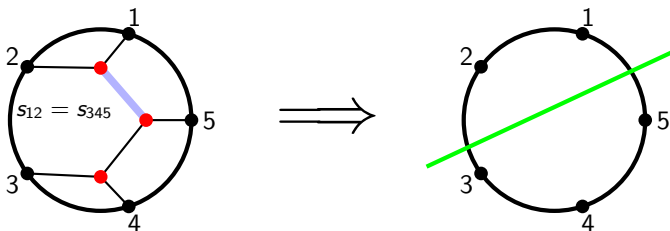
▷ **Helpful:** Such diagrams only have *planar kinematic invariants*.

Planar Kinematic Invariants & Chords on a Graph

▷ A planar kinematic invariant of length l is given by

$$s_{i,i+1,\dots,i+l-1} = \left(\sum_{p=0}^{l-1} k_{i+p} \right)^2, \quad (5)$$

with $2 \leq l \leq n-2$ and $1 \leq i \leq n$ is the particle label.



The Planar Scattering Potential

In any basis of kinematic invariants, the dimension is $\frac{n(n-3)}{2}$.

Idea $\implies$ Express the potential in terms of a *planar* basis.

$$SL(2, \mathbb{C}) \quad \checkmark \quad \implies \quad s_{ij} \rightarrow s_{ij} - m^2$$

The Planar Scattering Potential

Denoting $x_{ij} \equiv x_i - x_j$, the potential for n -particles is:

$$\mathcal{S}(\{x_i\}) = \sum_{k=2}^{n-2} \sum_{i=1}^k (s_{i, \dots, i+n-k-1} - m^2) \log \left(\frac{|x_{i, i+n-k-1}| |x_{i+n-k, i-1}|}{|x_{i, i+n-k}| |x_{i+n-k-1, i-1}|} \right)$$

Two Geometric Features

- ▷ **Count basis** using maximally symmetric Ferrer's shapes.
- ▷ $SL(2, \mathbb{C})$ invariance manifest due to emergence of **cross ratios**.

Building & Breaking Diagrams

The Geometry of Catalan Numbers

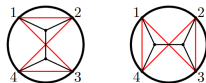
Combinatorics of Bi-adjoint ϕ^3 Diagrams - The Catalan Numbers

▷ The n^{th} Catalan number C_n is given by [33]

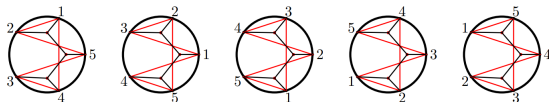
$$C_n = \frac{1}{n+1} \binom{2n}{n} \implies 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, \dots$$

Connection: Bi-adjoint ϕ^3 Diagrams & Catalan Numbers [12]

$$\text{Number of diagrams for } n\text{-particles} = C_{n-2}$$



$$n = 4 \longleftrightarrow C_2 = 2$$



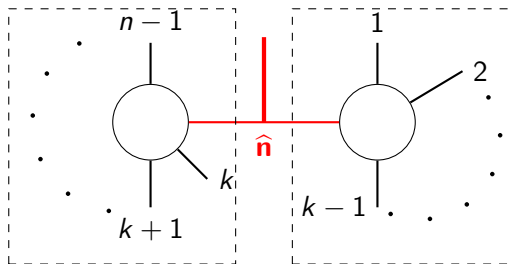
$$n = 5 \longleftrightarrow C_3 = 5$$

The Catalan Numbers & the Bi-adjoint ϕ^3 Theory

- ▷ One recursion relation that generates the Catalan numbers is *Segner's recurrence* [31],

$$C_{n-2} = \sum_{k=1}^{n-2} C_{k-1} C_{n-2-k}, \quad C_0 = 1. \quad (6)$$

Right-Hand-Side Diagram Factorisation



The $U(1)$ Decoupling Identity

The $U(1)$ Decoupling Identity

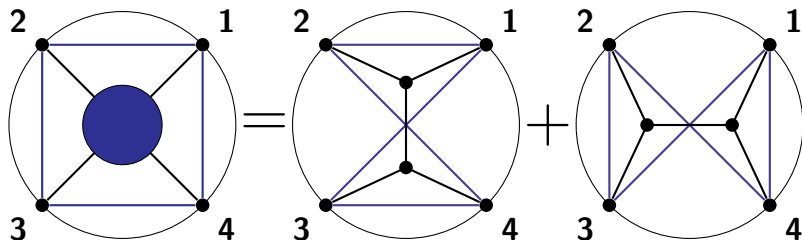
For an n -particle amplitude, the **sum** over all $(n - 1)$ **cyclic** permutations of external legs — where a **fixed leg remains in place** while the **others cycle** — must **vanish** [17, 24].

▷ Cycling through the n^{th} particle, simply gives

$$\sum_{k=1}^{n-1} A_n(1, 2, 3, \dots, k, n, k+1, \dots, n-1) = 0. \quad (7)$$

▷ Arises when one of the generators $T^a \propto \mathbf{1}$ [15].

Pictorial Representation for $n = 4$ Particles


2

 $m_4(1234|1234)$

=

1

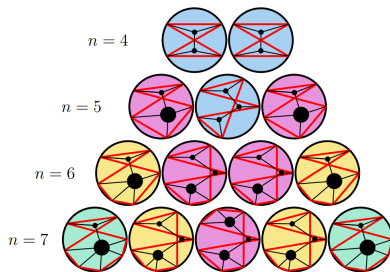
 $m_4(1234|1243)$

+

1

 $m_4(1234|1423)$

Counting the Diagrams



$m \backslash k$	1	2	3	4	5
2	1	1			
3	2	1	2		
4	5	2	2	5	
5	14	5	4	5	14

▷ Pictorial $U(1)$ decoupling factorisation.

▷ Combinatorial array of $U(1)$ decoupling factorisation $n = m + 2$.

The Narayana Numbers vs. Our Table

▷ C_n admit a *refinement* given by the Narayana numbers [29]

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}. \quad (8)$$

m \ k	1	2	3	4	5
1	1				
2	1	1			
3	1	3	1		
4	1	6	6	1	
5	1	10	20	10	1

m \ k	1	2	3	4	5
1	1				
2	1	1			
3	2	1	2		
4	5	2	2	5	
5	14	5	4	5	14

$$C_m = \sum_{k=1}^m N(m, k)$$

$$C_m = \sum_{k=1}^m C_{k-1} C_{m-k}$$

Emergent Correlation Functions from Colour-Ordered CHY

An Off-Shell Scattering Potential

Correlation Functions

LSZ Theorem: S-matrix $\leftrightarrow$ time-ordered correlation functions [23].

Fourier-Transformed n -point Green's Function

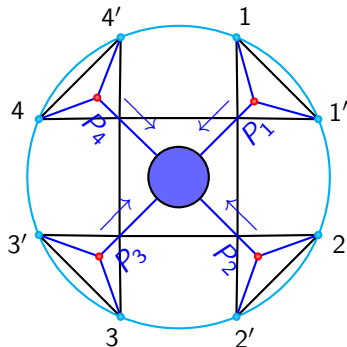
$$\mathcal{G}(P_1, \dots, P_n) = \int \prod_{a=1}^n dx_a e^{iP_a \cdot x_a} \langle \phi(x_1) \cdots \phi(x_n) \rangle \quad (9)$$

On-shell limit:

$$\mathcal{G}(P_1, \dots, P_n) \longrightarrow \left(\prod_{a=1}^n \frac{1}{P_a^2} \right) \mathcal{A}_n(P_1, \dots, P_n). \quad (10)$$

- ▷ $\mathcal{A}_n$ appears as the residue at the poles $P_a^2 = 0$.
- ▷ Off-shell terms in $\mathcal{G}$ do not contribute to the S-matrix.

A Specific Colour-Ordered Amplitude with Off-Shell Momenta



▷ Off-shell momentum

$$P_a^\mu \equiv p_a^\mu + \tau_a q^\mu, \quad (11)$$

with $p_a^2 = q^2 = 0$ & $0 \leq \tau_a \leq 1$.

▷ Start with the amplitude

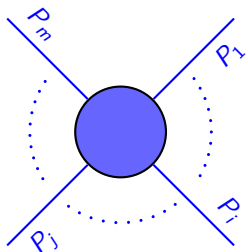
$$m_n(1, 1', \dots, m, m' | 1', 1, \dots, m', m)$$

▷ But, what about the complicated sub-diagram?

$$m_8(11'22'33'44' | 1'12'23'34'4)$$

Colour-Ordered Amplitudes $\implies$ Correlation Functions

- ▷ Complicated sub-diagram is an m -point correlator in disguise!


$$= \mathcal{G}(P_1, \dots, P_m)$$

- ▷ Explicitly computed for $2m = 6$ external particles using CHY.

The Off-Shell System

Physical Setup: The $2m$ particles live on a real interval $I = [0, 1]$, where the scattering potential governs their positions [13].

System Rules: Interaction strength between particles i & j is s_{ij}

- ▷ $s_{ab} \neq 0$ for $a \neq b \implies a$ & b **repel** each other.
- ▷ $s_{a'b'} = 0 \implies a'$ & b' **do not interact** among themselves.
- ▷ $s_{ab'} \neq 0 \implies$ Primed $\leftrightarrow$ Unprimed interactions permitted.

Consequences:

- ▷ Solve for positions of $m - 3$ dynamical unprimed particles.
- ▷ For $s_{ij} = 0$, particles coalesce without divergence, and the number of valid solutions changes.

An Off-Shell Potential

Number of Solutions to the Off-Shell Scattering Equations

$$\mathcal{N}_m = (m-3)!(m-2)^m \quad (12)$$

Off-Shell Scattering Potential

The scattering potential in terms of the off-shell momenta is

$$\begin{aligned} \mathcal{S}(x) = \sum_{a \neq b} \left(P_a \cdot P_b - \frac{\tau_a}{\tau_b} (P_b)^2 \right) \log |x_a - x_b| \\ + \sum_{a,b} \frac{\tau_a}{\tau_b} (P_b)^2 \log |x_{a'} - x_b|. \end{aligned} \quad (13)$$

Colour Ordering

Extra Slides

Colour Ordering #1

Goal

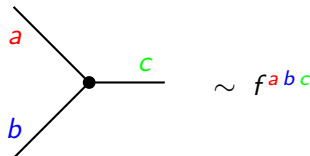
Organise amplitudes to separate colour and kinematics, revealing simpler analytic structure.

Example: Yang-Mills (set $g = 1$) [34, 28]

$$\mathcal{L} = -\frac{1}{4} \sum_a F_{\mu\nu}^a F^{a\mu\nu}, \quad F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc} A_\mu^b A_\nu^c \quad (14)$$

Feynman Rules

Interactions include f^{abc}



Colour Ordering #2

Interactions in Bi-adjoint ϕ^3 Theory

$$\mathcal{L}_{\text{int.}}^{\text{BA-}\phi^3} = -f_{abc}\tilde{f}_{\tilde{a}\tilde{b}\tilde{c}}\phi^{a\tilde{a}}\phi^{b\tilde{b}}\phi^{c\tilde{c}}. \quad (15)$$

▷ Use $U(N)$ generators as basis $\implies U(N) = SU(N) \times U(1)$:

$$i[T^a, T^b] = \sum_{c=1}^{N^2} f^{abc} T^c. \quad (16)$$

Colour Ordering #3

$$(T^a)_{ij} \longrightarrow \text{---} a \text{---} \begin{array}{c} \nearrow j \\ \searrow i \end{array}$$

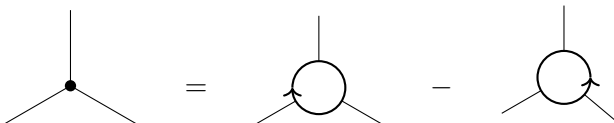
$$\begin{aligned} \text{Tr}(T^a T^b) &= (T^a)_{ij} (T^b)_{ji} = \text{---} a \text{---} \begin{array}{c} \nearrow j \\ \searrow i \end{array} \text{---} b \text{---} \\ &= \text{---} a \text{---} \text{---} b \text{---} \\ &= \delta_{ab} \end{aligned}$$

Colour Ordering #4

Strategy [14, 7]

Replace structure constants by generator traces

$$f^{abc} = \text{Tr}(T^a T^b T^c) - \text{Tr}(T^a T^c T^b).$$



Colour Decomposition #6 - Tree-Level Example #1

$$\begin{aligned}
 & \text{Diagram 1: A tree-level diagram with two vertices. The left vertex has incoming lines a_1 and a_2, and an outgoing line b. The right vertex has incoming line b, and outgoing lines a_3 and a_4. The diagram is labeled with $f^{a_1 a_2 b}$ and $f^{b a_3 a_4}$, and a double arrow indicates the contraction $f^{a_1 a_2 b} f^{b a_3 a_4}$.$$

$$= \left(\text{Diagram 2} - \text{Diagram 3} \right) \left(\text{Diagram 4} - \text{Diagram 5} \right)$$

$$= \left(\text{Tr}(T^{a_1} T^{a_2} T^b) - \text{Tr}(T^{a_1} T^b T^{a_2}) \right) \left(\text{Tr}(T^{a_3} T^{a_4} T^b) - \text{Tr}(T^{a_3} T^b T^{a_4}) \right)$$

$$= \text{Diagram 6} - \text{Diagram 7} - \text{Diagram 8} + \text{Diagram 9}$$

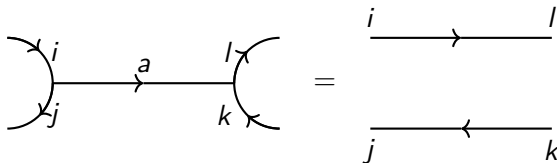
The diagrams are represented as follows:

- Diagram 2: A vertex with incoming lines \$a_1, a_2\$ and outgoing line \$b\$.
- Diagram 3: A vertex with incoming line \$b\$ and outgoing lines \$a_1, a_2\$.
- Diagram 4: A vertex with incoming line \$b\$ and outgoing lines \$a_3, a_4\$.
- Diagram 5: A vertex with incoming lines \$a_3, a_4\$ and outgoing line \$b\$.
- Diagram 6: Two vertices connected by a line \$b\$. The left vertex has incoming lines \$a_1, a_2\$ and the right vertex has outgoing lines \$a_3, a_4\$.
- Diagram 7: Two vertices connected by a line \$b\$. The left vertex has incoming lines \$a_1, a_2\$ and the right vertex has incoming lines \$a_3, a_4\$.
- Diagram 8: Two vertices connected by a line \$b\$. The left vertex has incoming lines \$a_1, a_2\$ and the right vertex has incoming lines \$a_3, a_4\$.
- Diagram 9: Two vertices connected by a line \$b\$. The left vertex has incoming lines \$a_1, a_2\$ and the right vertex has incoming lines \$a_3, a_4\$.

Colour Ordering #5 - Why $U(N)$? [7]

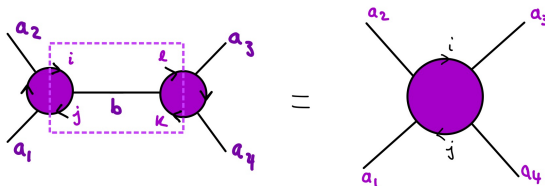
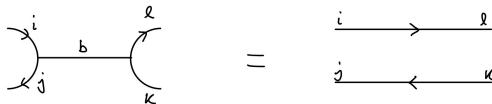
The $U(1)$ Completeness Relation

$$\sum_{a=1}^{N^2} (T^a)_{ij} (T^a)_{kl} = \delta_{il} \delta_{jk}, \quad (17)$$



Let's look at its importance with an example.

Colour Decomposition #7 - Tree-Level Example #2



$$\text{Tr}(T^{a_1} T^{a_2} T^b) \text{Tr}(T^b T^{a_3} T^{a_4}) = \text{Tr}(T^{a_1} T^{a_2} T^{a_3} T^{a_4})$$

Colour Ordering #8 - The Final Result

Result - Tree-level QCD amplitude [15, 14]

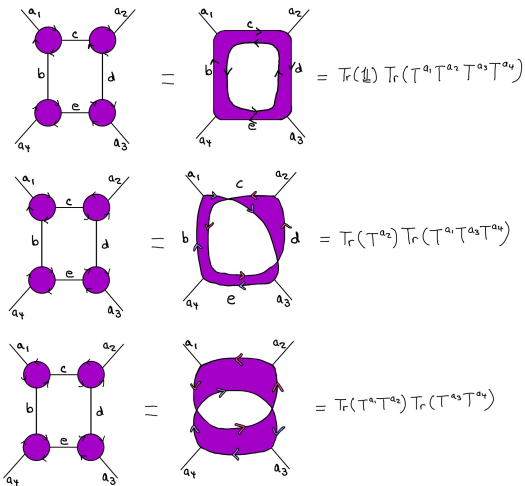
$$\mathcal{A}_n^{\text{tree}} = \sum_{\rho \in S_n / \mathbb{Z}_n} \text{Tr}(T^{a_{\rho(1)}} \dots T^{a_{\rho(n)}}) A_n(\rho(1), \dots, \rho(n)). \quad (18)$$

Colour Decomposition - One-Loop Example #1

$$\begin{aligned}
 & \text{Circular diagram with legs } a_1, a_2, a_3, a_4 \text{ and arcs } c, d, e, b \\
 &= \left(\text{Tree}(a_1, b, c) - \text{Tree}(a_1, c, b) \right) \left(\text{Tree}(a_2, c, d) - \text{Tree}(a_2, d, c) \right) \\
 & \quad \cdot \left(\text{Tree}(a_3, d, e) - \text{Tree}(a_3, e, d) \right) \left(\text{Tree}(a_4, e, b) - \text{Tree}(a_4, b, e) \right)
 \end{aligned}$$

$$\begin{aligned}
 &= \left(\text{Box}(a_1, b, c, d) - \text{Box}(a_1, c, b, d) \right) \\
 & \quad - \left(\text{Box}(a_1, c, d, e) - \text{Box}(a_1, d, c, e) \right) \\
 & \quad + \left(\text{Box}(a_1, d, e, b) - \text{Box}(a_1, e, d, b) \right) \\
 & \quad + (13 \text{ others})
 \end{aligned}$$

Colour Decomposition - One-Loop Example #2



CHY Amplitudes & A Colour-Ordered Scalar Theory

Adding Colour to CHY [11]

Colour-dressed integrands $\tilde{\mathcal{I}}_L$ & $\tilde{\mathcal{I}}_R$ for Yang-Mills ($s = 1$) & Einstein gravity ($s = 2$):

$$\tilde{\mathcal{I}}_L \tilde{\mathcal{I}}_R = \left(\sum_{\rho \in S_n / \mathbb{Z}_n} \frac{\text{Tr}(T^{a_{\rho(1)}} \dots T^{a_{\rho(n)}})}{(x_{\rho(1)} - x_{\rho(2)}) \dots (x_{\rho(n)} - x_{\rho(1)})} \right)^{2-s} (\text{Pf}' \Psi)^s. \quad (19)$$

But what about for $s = 0$?

$\implies$ A scalar theory with two independent colour-orderings?!

CHY Amplitude Calculation

Extra Slides

4-Point CHY Amplitude in Bi-adjoint ϕ^3 Theory

We compute the color-ordered partial amplitude:

$$m_4(1234|1234) = \frac{1}{\text{Vol}(\text{SL}(2, \mathbb{C}))} \int \prod_{a=1}^4 dx_a \prod_{a=1}^4 \delta(E_a) \text{PT}(1234) \text{PT}(1234),$$

with

$$E_a = \sum_{\substack{b=1 \\ b \neq a}}^4 \frac{s_{ab}}{x_a - x_b}, \quad \text{PT}(1234) = \frac{1}{(x_1 - x_2)(x_2 - x_3)(x_3 - x_4)(x_4 - x_1)}.$$

Gauge fixing:

$$x_2 = 0, \quad x_3 = 1, \quad x_4 = \infty.$$

This removes $\delta(E_2), \delta(E_3), \delta(E_4)$ and introduces the Fadeev–Popov like gauge fix:

$$(x_2 - x_3)^2 (x_3 - x_4)^2 (x_2 - x_4)^2.$$

Solving the Integral

After gauge fixing, the amplitude becomes:

$$A_4 = \int dx_1 \delta(E_1) \frac{(x_2 - x_3)^2 (x_3 - x_4)^2 (x_2 - x_4)^2}{(x_1 - x_2)^2 (x_2 - x_3)^2 (x_3 - x_4)^2 (x_4 - x_1)^2}.$$

With $x_2 = 0$, $x_3 = 1$, $x_4 = \infty$, this simplifies to:

$$A_4 = \int dx_1 \delta(E_1) \frac{1}{x_1^2}.$$

Scattering equation:

$$E_1 = \frac{s_{12}}{x_1} + \frac{s_{13}}{x_1 - 1} = 0 \quad \Rightarrow \quad x_1 = \frac{s_{12}}{s_{12} + s_{13}}.$$

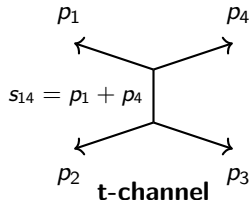
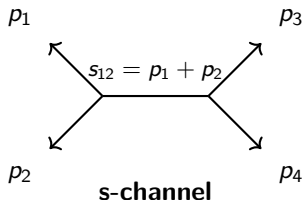
Final Result and Diagrammatic Interpretation

Plug into the integral:

$$A_4 = \frac{x_1(x_1 - 1)}{s_{12} + s_{13}} \cdot \frac{1}{x_1^2} = \frac{s_{13}}{s_{12}(s_{12} + s_{13})}.$$

Using Mandelstam variables:

$$s = s_{12}, \quad u = s_{13}, \quad s + t + u = 0, \quad \Rightarrow \quad A_4 = -\frac{u}{st} = \frac{1}{s} + \frac{1}{t}.$$



The Scattering Potential Picture

Extra Slides

The Positive Kinematic Region [13]

SL(2, $\mathbb{C}$) Gauge Fixing : $\{x_1 = 0, x_2 = 1, x_3 = \infty\}$

$$\begin{aligned}\mathcal{S}(x) &= \sum_{1 \leq a < b \leq n} s_{ab} \log |x_a - x_b| \\ &= \sum_{4 \leq a < b \leq n} s_{ab} \log |x_a - x_b| + \sum_{a=3}^n s_{a1} \log |x_a| + \sum_{a=3}^n s_{a2} \log |x_a - 1|\end{aligned}$$

A positive basis

$$\mathcal{K}_n^+ = \begin{cases} s_{ab} > 0 \\ s_{a1} > 0 \\ s_{a2} > 0 \end{cases} \qquad \dim \mathcal{K}_n^+ = \frac{n(n-3)}{2}$$

Interpretation as a Potential [13]

In $\mathcal{K}_n^+$, the scattering equations have $(n - 3)!$ **real** solutions.

$$\mathcal{S}(x) = \sum_{4 \leq a < b \leq n} \underbrace{s_{ab}}_{\substack{\text{interaction strength} \\ a \leftrightarrow b}} \log |x_a - x_b| + \sum_{a=3}^n \underbrace{s_{a1}}_{\substack{\text{interaction strength} \\ a \leftrightarrow 1}} \log |x_a| + \sum_{a=3}^n \underbrace{s_{a2}}_{\substack{\text{interaction strength} \\ a \leftrightarrow 2}} \log |x_a - 1|$$

$\mathcal{S}(x) \leftrightarrow -V(x)$

$$\frac{\partial V}{\partial x_a} = 0 \Rightarrow \text{scattering equations}$$

Solutions & Particles on a Real Line [13]

$V(x)$: the potential for $n-3$ particles moving in the real line $I = [0, 1]$.

$$-V(x) = \sum_{4 \leq a < b \leq n} s_{ab} \log |x_a - x_b| + \sum_{a=3}^n s_{a1} \log |x_a| + \sum_{a=3}^n s_{a2} \log |x_a - 1|$$

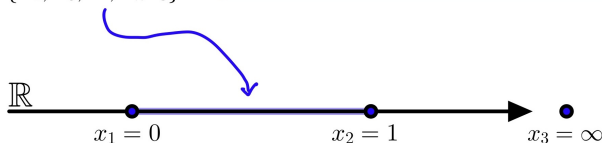
Solution = Equilibrium State

#of Solutions = #of Equilibrium States

= # of ways to permute the $n-3$ particles in $I = [0, 1]$

= $(n-3)!$

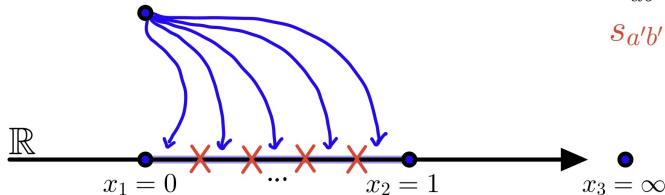
$\{x_4, x_5, \dots, x_{n-3}\}$



$x_i \rightarrow 0 \Rightarrow V(x) \rightarrow \infty$	$x_i \rightarrow x_j \Rightarrow V(x) \rightarrow \infty$	$x_i \rightarrow 1 \Rightarrow V(x) \rightarrow \infty$	$x_i \rightarrow \pm\infty \Rightarrow V(x) \rightarrow \infty$
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The Off-Shell Case

$\{\cancel{1}, \cancel{2}, \cancel{3}, \dots, m, \underline{1'}, 2', 3', \dots, m'}\}$



$$s_{ab} \neq 0$$

$$s_{ab'} \neq 0$$

$$s_{a'b'} = 0$$

$x_i \rightarrow 0 \Rightarrow V(x) \rightarrow \infty$	$x_i \rightarrow x_j \Rightarrow V(x) \rightarrow \infty$	$x_i \rightarrow 1 \Rightarrow V(x) \rightarrow \infty$	$x_i \rightarrow \pm\infty \Rightarrow V(x) \rightarrow \infty$
	$x_i \rightarrow x_{j'} \Rightarrow V(x) \rightarrow \infty$	$x_{i'} \rightarrow x_j \Rightarrow V(x) \rightarrow \infty$	

$$\begin{aligned} \# \text{of Solutions} &= \# \text{of Equilibrium States} \\ &= (m-3)!(m-2)^m \end{aligned}$$

A Taste of Twistor Geometry in 4d [27]

In 4d, *twistor space* $\mathbb{PT} \subseteq \mathbb{CP}^3$

▷ $Z^A = (Z^1, \dots, Z^4) \neq (0, \dots, 0)$, where $rZ^A \sim Z^A \quad \forall r \in \mathbb{C}^*$.

Z^A can be organised into two **two-dimensional Weyl** spinors:

$$Z^A = (\mu^{\dot{\alpha}}, \lambda_{\alpha}).$$

Complexified Minkowski space $\mathbb{M}_{\mathbb{C}}$ is related to twistor space via the *incidence relations*

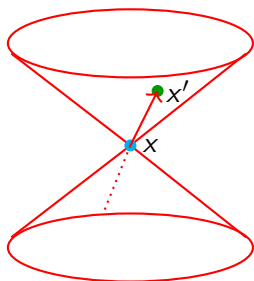
$$\mu^{\dot{\alpha}} = x^{\alpha\dot{\alpha}} \lambda_{\alpha}. \tag{20}$$

For fixed x , these define a projective line $\mathbb{CP}^1 \subset \mathbb{PT}$.

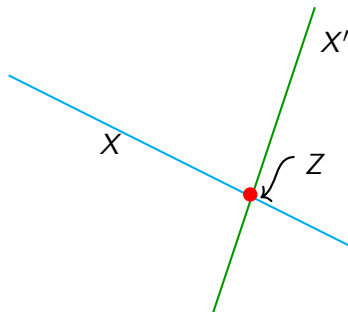
This is a ***non-local correspondence***.

The Twistor Correspondence in 4d [27, 1]

Space-time $\mathbb{M}_{\mathbb{C}}$



Twistor Space $\mathbb{PT}$

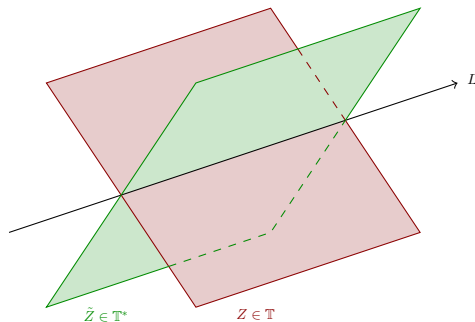


What does Z^A correspond to in $\mathbb{M}_{\mathbb{C}}$?

A fixed $Z \in \mathbb{PT} \longleftrightarrow$ totally null 2-plane on $\mathbb{M}_{\mathbb{C}} \equiv \alpha$ -plane.

Connection Between $\mathbb{PA}$ and $\mathbb{PT}$ [18]

In 4d, $\mathbb{PA} = T^*\mathbb{PT}$ and $\mathbb{PA} = T^*\mathbb{PT}^*$ - ambidextrous!



▷ L is a *null line* in $\mathbb{M}_{\mathbb{C}}$ at $Z \cdot \tilde{Z} = 0$.

$$\mathbb{PA} = \{(Z, \tilde{Z}) | Z \cdot \tilde{Z} = 0\}$$